NARAYANA
IIT - JEE ACADEMY - INDIA

Sec: SR STAR - A & APEX, ALL IITS | DT: 28/03/2026
Time: 3 Hrs | GTM – 07 | Max. Marks: 300

KEY SHEET

PHYSICS

01	02	03	04	05	06	07	08	09	10
D	D	C	C	B	B	C	C	D	C
11	12	13	14	15	16	17	18	19	20
A	D	B	A	D	A	A	C	D	B
21	22	23	24	25					
20	110	4	50	5					

CHEMISTRY

26	27	28	29	30	31	32	33	34	35
C	C	C	B	B	B	A	A	C	A
36	37	38	39	40	41	42	43	44	45
D	D	C	A	D	D	A	A	D	A
46	47	48	49	50					
3	22	64	6	1					

MATHEMATICS

51	52	53	54	55	56	57	58	59	60
C	D	A	B	D	B	A	D	B	C
61	62	63	64	65	66	67	68	69	70
B	D	D	C	C	C	C	B	B	C
71	72	73	74	75					
120	3759	28	3	33					

Rankers Academy JEE

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HINTS
PHYSICS

1.

$$E_n = \frac{q^2B^2r_n^2}{2m}$$

$$\because\ mv_n r_n = n\hbar \text{ and } r_n = \frac{mv_n}{qB}$$

$$\Rightarrow r_n = \frac{m}{qB}\frac{n\hbar}{mr_n} \Rightarrow r_n^2 = \frac{n\hbar}{qB}$$

$$\Rightarrow E_n = \frac{q^2B^2r_n^2}{2m} = \frac{q^2B^2}{2m} \times \frac{n\hbar}{qB} = n\left(\frac{qBh}{4\pi m}\right)$$

2. The output D for the given combination $D = \overline{(A+B).C} = \overline{(A+B)} + \overline{C}$

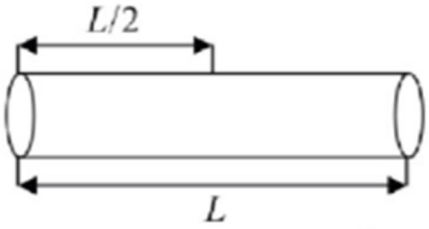
If $A = B = C = 0$ then $D = \overline{(0+0)} + \overline{0} = \overline{0} + \overline{0} = 1+1 = 1$

If $A = B = 1, C = 0$ then $D = \overline{(1+1)} + \overline{0} = \overline{1} + \overline{0} = 0+1 = 1$

3. Same intensity since $I_{min} = 0$
S_1, S_2 must not have equal phase but same phase.

As $\phi_1 = \frac{\pi}{2}, \phi_2 = \frac{7\pi}{3}$

4. The fundamental frequency in organ pipe with both end open is $f = \frac{v}{2L}$

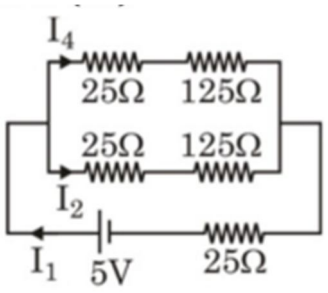


With additional rate at $\frac{L}{2}$, the fundamental frequency becomes

$$f' = \frac{v}{2L} = \frac{v}{\frac{2L}{2}} = \frac{340\,m/s}{29\times10^{-2}\,m} = 1172\,Hz$$

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5.

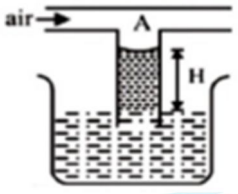


$$R_{eq} = \frac{200 \times 200}{400} + 25 = 125\Omega$$

$$I_2 = I_4 = \frac{I_1}{2}$$

$$\frac{I_1}{I_2} = 2$$

6.



According to Bernoulli equation,

$$P_1 + \frac{1}{2}\rho v^2 + \rho g h = \text{constant} \qquad (i)$$

If velocity of air will increase the pressure at point (A) must be decreases to maintain equation (i).

According to question, air will start to blown through the horizontal tube so pressure point (A) will decrease.

∴ The height of liquid rise in capillary tube will increases.

7.

$$f = F\cos 37°$$

$$\mu N = \frac{4F}{5}$$

$$\mu\left(Mg - \frac{3F}{5}\right) = \frac{4F}{5}$$

$$\frac{Mg}{5} - \frac{3F}{25} = \frac{4F}{5}$$

$$F = 5 \times \frac{400}{23}$$

8. Let E be the e.m.f. and r be the internal resistance of battery. Then,

$$0.5 = \frac{E}{10 + r}$$

$$\Rightarrow \quad E = 5 + 0.5r$$

and $\quad 0.8 = \frac{E}{5 + r}$

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$\Rightarrow \quad r = \frac{10}{3}\Omega$ and $E = \frac{20}{3}\Omega$

9.

$$AgCN(s) \rightleftharpoons Ag^{+}(aq.) + CN^{-}(aq.) \qquad K_{eq} = K_{sp}$$

$$CN^{-}(aq.) + H^{+}(aq.) \rightleftharpoons HCN(aq.) \qquad K_{eq} = \frac{1}{ka}$$

$$AgCN(s) + H^{+}(aq.) \rightleftharpoons Ag^{+}(aq.) + HCN(aq.)$$

$$K_{eq} = \frac{K_{sp}}{K_a}$$

$$K_{eq} = \frac{[Ag^{+}][HCN]}{[H^{+}]} = \frac{K_{sp}}{K_a}$$

$$\frac{s^2}{10^{-4}} = \frac{1.6 \times 10^{-16}}{6.4 \times 10^{-10}}$$

$$s^2 = \frac{1}{4} \times 10^{-10}$$

$$s = \frac{1}{2} \times 10^{-5}$$

$$s = 5 \times 10^{-6}$$

10. Given Q = 400J

$$Q = nC_p\Delta T$$

$$= n\left(\frac{5}{2}R\right)\Delta T$$

$$= \frac{5}{2}nR\Delta T$$

For constant pressure

$$W = nR\Delta T$$

$$W = \frac{2}{5} \times 400$$

$$= 160J$$

11. EMF induced in the triangular loop is given as loop is given as $e = \frac{d\phi}{dt}$

$$\Rightarrow e = A\frac{dB}{dt} \Rightarrow e = \sqrt{3} \times \sqrt{3} = 3V$$

By symmetry in each rod emf will be IV. The current in the triangular loop is given as

$$i = \frac{e}{R} = \frac{3}{5} = 0.6A$$

Voltage drop between A and B $= 0.6 \times 1 = 0.6V$

So voltage across AB is given as

$V_{AB} = 1 - 0.6 = 0.4V$

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12. The output of both AND gates will be zero. So, OR gate does not respond.

13. $\frac{\lambda}{4} = \ell_1 + e \quad ...(1)$

$\frac{3\lambda}{4} = \ell_2 + e \quad ...(2)$

From (1) and (2)

$e = 2cm$

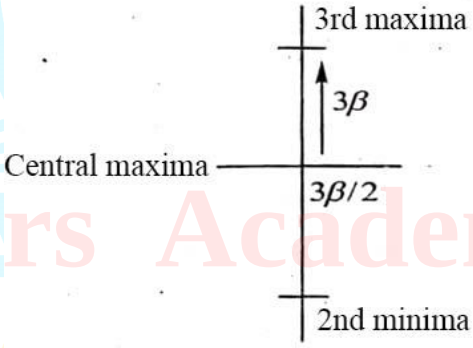
14. $B_0 = \frac{E_0}{C} = \frac{300}{3x10^8} = 10^{-6}T$

$F_{max} = qvB_0 = 1.6x\,10^{-19} x\,2x10^7\,x10^{-6} = 3.2x10^{-18}N$

15. d=0.5mm and D=0.5m

Separation = $3\beta + 1.5\beta = 4.5\beta$

$= 4.5 \times \frac{\lambda D}{d} = 2.25mm$



16. $V=u^aA^bt^c$

Dimensionally

$L^3 = (LT^{-1})^a (L^2)^b (T)^c$

$\Rightarrow a + 2b = 3$

$c - a = 0$

17. v_o const

$\Rightarrow y = V_0 t$

$V_x = Ay$

$dx = AV_0 t\,dt$

$\Rightarrow x = \frac{AV_0}{2} t^2 \qquad \Rightarrow x = \frac{AV_0}{2} \frac{y^2}{V_0^2}$

$\Rightarrow x = \frac{Ay^2}{2V_0}$

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18. The body loses contact at angle 'θ' with vertical given by $\cos\theta = \frac{2}{3}$

$\Rightarrow$ Tangential acceleration $a = g\sin\theta = \frac{\sqrt{5}g}{3}$

19. The time constant of the CR circuit is $\pi = CR = 10^{-3}$ s. then, we have

$q(t) = CV[1-\exp(-t/\pi)]$

$= 210^{-9}[1-\exp(-t/10^{-3})]$

The electric field in between the plates at time t is

$E = \frac{q(t)}{\varepsilon_0 A} = \frac{q}{\pi\varepsilon_0}; A = \pi[1]^2\,\text{m}^2 = \text{area of the plates.}$

Consider now a circular loop of radius (1/2) m parallel to the plates passing through P. The magnetic field B at all points on the loop is along the loop and of the same value.

The flux Φ_E through this loop is

The flux $\Phi_E = \pi \times \left(\frac{1}{2}\right)^2 = \frac{\pi E}{4} = \frac{q}{4\varepsilon_0}$

The displacement current

$i_d = \varepsilon_0 \frac{d\Phi_E}{dt} = \frac{1}{4}\frac{dq}{dt} = 0.5\times10^{-6}e^{-1}$

At $t = 10^{-3}s$. Now, applying ampere-maxwell law to the loop. We get

$B \times 2\pi \times \left(\frac{1}{2}\right) = \mu_0(i_d) = 0.5\times10^{-6}\,\text{e}^{-1}$

$B = 0.74\times10^{-13}T$

20. $\frac{m(\sqrt{3})+(n-1)m7\sqrt{3}}{nm} = \frac{u^2\sin2\theta}{g}$

$\Rightarrow \frac{\sqrt{3}+(n-1)7\sqrt{3}}{n} = \frac{100}{10}\times\frac{\sqrt{3}}{2}$

$\Rightarrow 1+(n-1)7 = 5n$

$1+7n-7 = 5n$

$2n = 6 \qquad n = 3$

21. Constrained Equation

$-2T_1 \times a_2 + 3T_1 \times a_1 = 0$

$a_2 = \frac{3}{2}a_1$

$3T_1 - m_1 g = m_1 a_1$

$3T_1 - 50 = 5a_1 \ldots\ldots\ldots 1$

$m_2 g - 2T_1 = m_2 a_2$

$20 - 2T_1 = 2a_2 = 3a_1 \ldots\ldots\ldots 2$

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Solving Eq. 1 & Eq. 2

We get

$-40 = 19a_1$

$a_1 = -\frac{40}{19} = \frac{2\times 20}{20-1}$

$n = 20$

22.

$I = \frac{2}{5}mr^2 + md^2$

$= \frac{2}{5}m5^2 + m10^2$

$= 110m$

$I = mk^2 = 110m$

$k^2 = 110$

$k = \sqrt{110}$

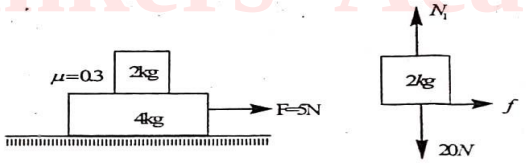
23. Width of central bright fringe = $\frac{2\lambda D}{d} = \frac{2\times 500\times 10^{-9}\times 80\times 10^{-2}}{0.20\times 10^{-3}}$

= 4 mm

24. For coil R = 200/1 = 200 Ω

With a.c. $i = \frac{200}{\sqrt{R^2 + x_L^2}}$

25.



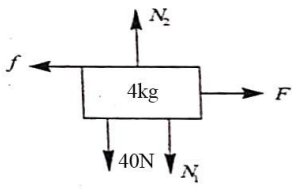
Limiting friction between blocks is $f_{max} = \mu N_1 = 0.3\times 20 = 6N$

To start sliding between blocks we use

$f_{max} = 2\left(\frac{F}{6}\right) = 6 \Rightarrow F = 18N$

$\Rightarrow$at F = 15N both blocks move together at $a = \frac{F}{6} = \frac{5}{2}m/s^2$

$\Rightarrow f = 2a = 2\times\frac{5}{2} = 5N$



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26. $\Delta G^\circ = -RT\ln K, T_1 = 25^\circ C$

$K_1 = 10$

ΔG° at $T_1 = 25^\circ C$

$T_1 = -8.314 \times 298 \times 2.303 \times \log 10 = -5.7 \text{kJ/mol}$

ΔG° at $T_2(100^\circ C) = -8.314 \times 373$

$\times 2.303 \times \log(100)$

$= -14.29 \text{kJ/mol}$

$\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ$

$\Rightarrow -5.71 = \Delta H^\circ - 298(\Delta S^\circ)$

and $-14.29 = \Delta H^\circ - 373(\Delta S^\circ)$

Hence,

At T_1

$\Delta G^\circ = -5.71$ kJ/mol

$\Delta H^\circ = 28.4$ kJ/mol

At T_2

$\Delta G^\circ = -14.29$ kJ/mol

$\Delta H^\circ = 28.4$ kJ/mol

27. Given, density of commercial acid =1.46 g/mL

So, mass of commercial acid = Density × Volume

$=1.46 \times 1000 = 1460$ g

$\therefore$ Mass of given 35 % HCl $= \frac{35}{100} \times 1460 = 511g$

Molarity of solution $= \frac{\text{Number of moles of acid}}{\text{Volume of acid (inlitre)}}$

$= \frac{\text{Weight of acid}}{\text{M.M of acid} \times \text{Volume of}} \text{acid (in litre)}$

$= \frac{511}{36.5 \times 1L} = 14M$

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28. (i) Rate of the reaction, $A_2 \underset{k_{-1}}{\overset{k_1}{\rightleftharpoons}} 2A$

$$r = r_{\text{forward}} - r_{\text{backward}}$$

$$= k_1[A_2] - k_{-1}[A]^2 \qquad \ldots(\text{i})$$

(ii) Also differential, rate of the reaction can be expressed as,

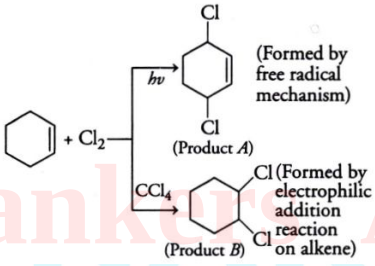
$$r = -\frac{d[A_2]}{dt} = +\frac{1}{2}\frac{d[A]}{dt}$$

So, the rate of appearance of A, i.e. $\frac{d[A]}{dt} = 2r$

$$= 2k_1[A_2] - 2k_{-1}[A]^2 \text{ (From Eq. (i))}$$

29. Pentacyanomangnate (III) is the IUPAC name of complex $\left[Mn(CN)_5\right]^{2-}$.

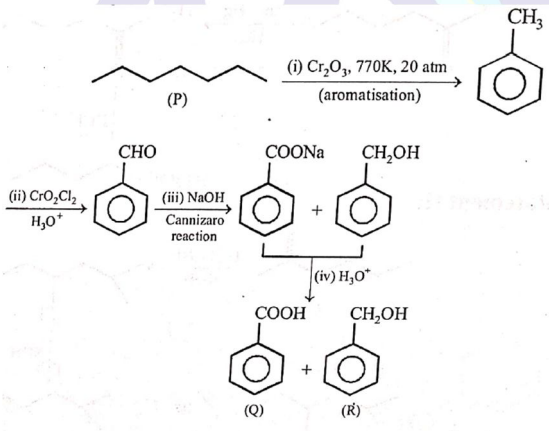
30. The complete reaction is as follow



Hence, option (B) is correct.

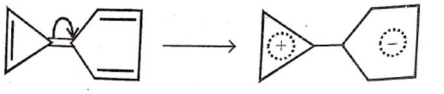
31. As ligand field increases, radiation of high energy is absorbed. Energy of radiation and wave number is the/directly proportional. Therefore strong field ligand absorb light of higher wave number. Order of strength of ligands $H_2O < NH_3 < CN^-$ *and also* $\Delta_t < \Delta_O$

32.

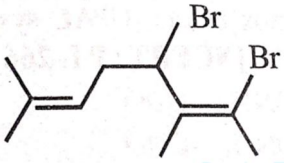


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33. Among the given compounds, the following compound has the highest dipole moment because both the +ve and –ve ends acquire aromaticity.



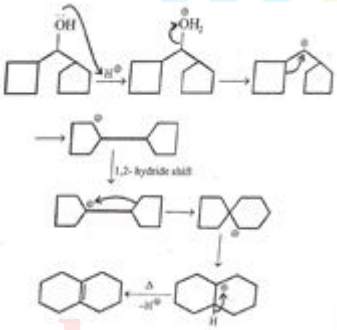
34.



There are two stereo – centers (n = 2)

So, No. of stereoisomer = $2^n = 2^2 = 4$

35.

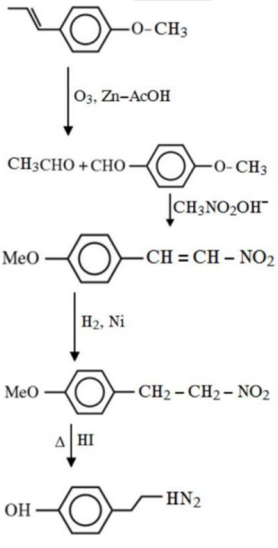


36. Dipole moment depends on both charge developed and bond lengths.

The dipole moment of $CH_3 - F$ is 1.847D and that of $CH_3 - Cl$ is 1.860 D.

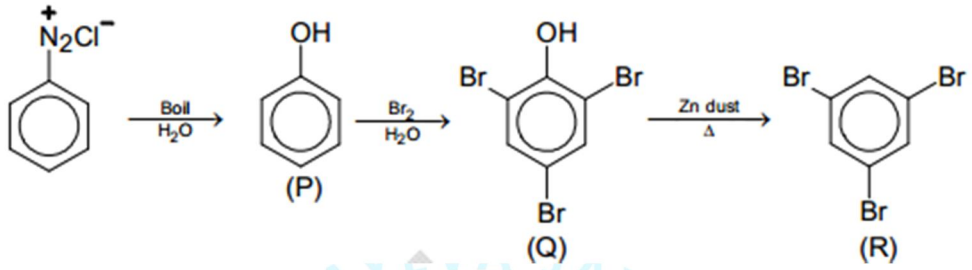
(NCERT data)

37.



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38.



Q is polar while R is non-polar.

$\therefore \quad \mu_R < \mu_Q$

39.

40. Correct order of ionic radii : $O^{2-} > Mg^{2+}$

This is because among isoelectronic species, the size of anions are greater than the size of cations.

Statement (II) is correct as both O^{2-} and Mg^{2+} are isoelectronic.

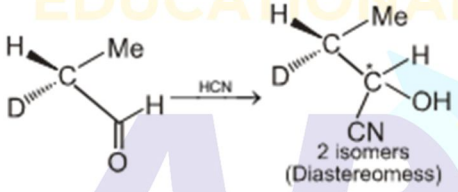
41. A is Aniline,

B is 2,4,6-tribromoaniline,

C is 1,3,5-tribromobenzene.

42.

43.



44. NCERT – XI, VOL – I, P.NO: 228, Q.NO: 7.45

$\left[H^+\right]$ from H_2S is negligible when compared with, $\left[H^+\right]$ from HCl

$$K_{a1} = \frac{\left[H^+\right]\left[HS^-\right]}{\left[H_2S\right]} \Rightarrow \left[HS^-\right] = K_{a1}\frac{\left[H_2S\right]}{\left[H^+\right]} = 9.1\times10^{-18}\times\frac{10^{-1}}{10^{-3}}$$

$$= 9.1\times10^{-6}\,M$$

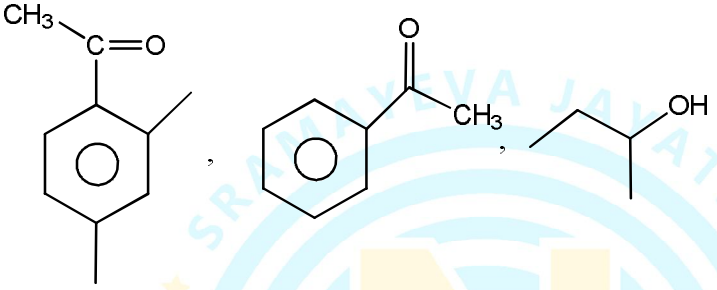
$$K_{a1}K_{a2} = \frac{\left[H^+\right]^2\left[S^{2-}\right]}{\left[H_2S\right]}\left[S^{2-}\right] = \frac{9.1\times10^{-8}\times1.2\times10^{-13}\times0.1}{\left(10^{-3}\right)^2}$$

$$= 10.92\times10^{-16}\,M$$

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45. Both V^{2+}, Co^{2+} possess same number of unpaired electrons.

46. Haloform test is given by molecules with $-\overset{\overset{O}{\|}}{C}-CH_3$ and $-\overset{\overset{OH}{|}}{CH}-CH_3$ group present in them thus total three compounds with this functional group will give haloform test i.e.,



47. Mass percent of alcohol

$$= \frac{\text{Mass of ethyl alcohol}}{\text{Total mass of solution}} \times 100$$

$$= \frac{1 \times 46}{1 \times 46 + (9 \times 18)} \times 100$$

$$= \frac{4600}{208}$$

$$= 22.11$$

Nearest integer = 22

48. For a 'n'th order reaction,

$$t_{\frac{1}{2}} \propto \frac{1}{a^{n-1}} \text{ where a = initial conc.}$$

$$\left(\frac{5 \times 10^{-3}}{25 \times 10^{-4}}\right)^{n-1} = \frac{8}{1} \Rightarrow n = 4$$

$$\frac{t_{1/2}}{1} = \left(\frac{5 \times 10^{-3}}{1.25 \times 10^{-3}}\right)^{n-1}$$

$$t_{1/2} = 64 \text{ hr}$$

49. With $C \equiv CH \rightarrow 1$ mole G.R. reacts

With $-OH \rightarrow 1$ mole G.R. reacts

With $CN \rightarrow 1$ mole G.R. reacts

With $CHO \rightarrow 1$ mole G.R. reacts

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With $COOCH_3 \rightarrow 2$ mole of G.R. reacts

Thus, total 6 moles of G.R. are consumed.

50. $60 \times x = 15 \times 0.02 \times 2$

$\therefore \quad x = 0.01M$

MATHEMATICS

51.

$$a_r = \left(cis\frac{2\pi}{9} \right)^r \text{ or } e^{i\frac{2\pi r}{9}} \quad r = 1,2,3,\cdots$$

$\therefore a_1, a_2, \ldots\ldots \text{are in GP}$

$$\begin{vmatrix} a_1 & a_2 & a_3 \\ a_4 & a_5 & a_6 \\ a_7 & a_8 & a_9 \end{vmatrix} = \begin{vmatrix} a_1 & a_1^2 & a_1^3 \\ a_1^4 & a_1^5 & a_1^6 \\ a_1^7 & a_1^8 & a_1^9 \end{vmatrix}$$

$$= a_1 \cdot a_1^4 \cdot a_1^7 \begin{vmatrix} 1 & a_1 & a_1^2 \\ 1 & a_1 & a_1^2 \\ 1 & a_1 & a_1^2 \end{vmatrix} = 0$$

$$\text{Now} \quad a_1 a_9 - a_3 a_7 = a_1^{10} - a_1^{10} = 0$$

52. P.Q divides AB in the ratio of 1 : 2 and 2 : 1 respectively and hence the co-ordinates of P and Q are

$$\left[\frac{1\times 2 + 2(-3)}{3}, \frac{1\times 3 + 2\times 2}{3} \right] \text{and} \left[\frac{2\times 2 + 1(-3)}{3}, \frac{2\times 3 + 1\times 2}{3} \right]$$

$$\Rightarrow \quad P\left(-\frac{4}{3}, \frac{7}{3} \right) \text{ and } Q\left(\frac{1}{3}, \frac{8}{3} \right) \text{ and } PQ = \frac{\sqrt{26}}{3}$$

Let the co-ordinates of R be (x,y)

Then, $QR = PQ \Rightarrow \left(x - \frac{1}{3} \right)^2 + \left(y - \frac{8}{3} \right)^2 = \frac{26}{9}$ …(i)

And QR is perpendicular to PQ

$$\Rightarrow \quad \frac{y - \frac{8}{3}}{x - \frac{1}{3}} \times \frac{\frac{8}{3} - \frac{7}{3}}{\frac{1}{3} + \frac{4}{3}} = -1 \quad \Rightarrow \quad \left(y - \frac{8}{3} \right) = -5\left(x - \frac{1}{3} \right) \quad \text{…(ii)}$$

From equation, (i) and (ii), we get

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$26\left(x-\frac{1}{3}\right)^2 = 26/9 \Rightarrow -\frac{1}{3} = \pm\frac{1}{3} \quad \Rightarrow \quad x = 0 \text{ or } x = 2/3$

From equation (ii), when $x = 0, y = 13/3$ and when $x = 2/3, y = 1$

So, the required co-ordinates of R are $\left(0, \frac{13}{3}\right)$ or $\left(\frac{2}{3}, 1\right)$

53. $D < 0$

$\Rightarrow \quad \left(-2(1+3m)\right)^2 - 4\left(1+m^2\right)(1+8m) < 0$

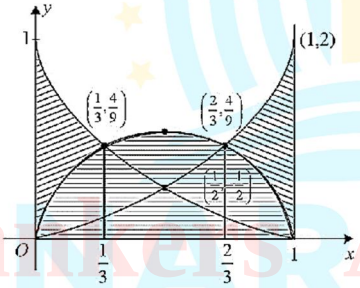
$\Rightarrow \quad 4\left[(1+3m)^2 - \left(1+m^2\right)(1+8m)\right] < 0$

$\Rightarrow \quad \left[\left(1+9m^2+6m\right) - \left(1+m^2+8m+8m^3\right)\right] < 0$

$\Rightarrow \quad 8m^3+8m^2-2m < 0 \quad \Rightarrow \quad (-2m)\left(4m^2-4m+1\right) < 0$

$\Rightarrow \quad m(2m-1)^2 > 0 \quad \Rightarrow \quad m > 0 \text{ and } m \neq \frac{1}{2} \quad \Rightarrow \quad m \in \left(0, \frac{1}{2}\right) \cup \left(\frac{1}{2}, \infty\right)$

54.



Let $C_1 : y = x^2; C_2 : y = (1-x)^2; C_3 : y = 2x(1-x)$. Then C_2 and C_3 intersect in (1/3,4/9) and C_1 intersect C_3 in (2/3,4/9).

Now $f(x) = \begin{cases} (1-x)^2 & for \quad 0 \le x \le 1/3 \\ 2x(1-x) & for \quad 1/3 \le x \le 2/3 \\ x^2 & for \quad 2/3 \le x \le 1 \end{cases}$

Required area (shaded portion)

$$= \int_0^{1/3} (1-x)^2\,dx + \int_{1/3}^{2/3} 2x(1-x)dx + \int_{2/3}^{1} x^2 dx = -\frac{1}{3}\left[(1-x)^3\right]_0^{1/3} + \left[x^2\right]_{1/3}^{2/3} - \frac{2}{3}\left[x^3\right]_{1/3}^{2/3} + \frac{1}{3}\left[x^3\right]_{2/3}^{1}$$

$$= -\frac{1}{3}\left[\frac{8}{27}-1\right] + \left(\frac{4}{9}-\frac{1}{9}\right) - \frac{2}{3}\left(\frac{8}{28}-\frac{1}{27}\right) + \frac{1}{3}\left(1-\frac{8}{27}\right)$$

$$= \frac{19}{81} + \frac{3}{9} - \frac{14}{81} + \frac{19}{81} = \frac{19+27-14+19}{81} = \frac{51}{81} = \frac{17}{27}$$

55. $2be = 6$

$be = 3$

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$$b\left(\frac{3}{5}\right)=3\Rightarrow b=5$$

$$\because a^2=b^2-b^2e^2$$

$$25-9=16$$

$$L.R.=\frac{2a^2}{b}=\frac{32}{5}$$

56. $$f(x)=\frac{\tan^{-1}\left(1-\{x\}\right)\cos^{-1}\left(1-\{x\}\right)\sin^{-1}\left(1-\{x\}\right)}{\sqrt{2\{x\}}\left(1-\{x\}\right)^2}$$

$$\therefore \quad P=\lim_{x\to 0^+} f(x)=\lim_{x\to 0^+}\frac{\tan^{-1}\left(1-x\right)\cos^{-1}\left(1-x\right)\sin^{-1}\left(1-x\right)}{\sqrt{2x}\left(1-x\right)^2}$$

$$\left(\because x\to 0^+\Rightarrow\{x\}=x\right)$$

$$=\lim_{x\to 0^+}\frac{\tan^{-1}}{(1-x)}\frac{\sin^{-1}(1-x)}{(1-x)}\frac{\cos^{-1}(1-x)}{\sqrt{2x}}$$

$$=\frac{\pi}{4}.\frac{\pi}{2}\times\lim_{x\to 0^+}\frac{\cos^{-1}(1-x)}{\sqrt{2}\sqrt{x}}=\frac{\pi^2}{8}\times\lim_{x\to 0^+}\frac{\theta}{\sqrt{1-\cos\theta}}$$

$$\left(\begin{array}{l}\text{Assuming } \cos^{-1}(1-x)=\theta\\ \text{As } x\to 0^+, 1-x\in(0,1)\\ \Rightarrow\theta\in(0,\pi/2)\\ \Rightarrow\sin\theta>0\end{array}\right)$$

$$=\frac{\pi^2}{8\sqrt{2}}\times\lim_{x\to 0^+}\frac{\theta}{\sqrt{2\sin^2\theta/2}}=\frac{\pi^2}{8\sqrt{2}}\times\lim_{\theta\to 0^+}\frac{\theta}{\sqrt{2}\left|\sin(\theta/2)\right|}$$

$$=\frac{\pi^2}{16}\lim_{\theta\to 0^+}\frac{\theta}{\sin\theta/2}\qquad\left(\because\sin\left(\frac{\theta}{2}\right)>0\right)$$

$$=\frac{\pi^2}{16}\lim_{\theta\to 0^+}2\frac{(\theta/2)}{\sin\theta/2}=\frac{\pi^2}{8}(1)=\frac{\pi^2}{8}$$

57. $$\Delta=\begin{vmatrix}x^2+x & x+1 & 2-2\\ 2x^2+3x-1 & 3x & 3x-3\\ 2x^2+2x+3 & 2x-1 & 2x-1\end{vmatrix}$$

$$\Delta=\begin{vmatrix}x^2+x & x+1 & x-2\\ -4 & 0 & 0\\ x^2+2x+3 & 2x-1 & 2x-1\end{vmatrix}\left[by\ R_2\to R_2-(R_1+R_3)\right]$$

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$$\Delta=\begin{vmatrix} x & x+1 & x-2 \\ -4 & 0 & 0 \\ 2x+3 & 2x-1 & 2x-1 \end{vmatrix}\left[by\ R_1\to R_1+\left(\frac{x^2}{4}\right)R_2 \text{ and } R_3\to R_3+\frac{x^2}{4}R_2\right]$$

$$\Delta=\begin{vmatrix} x & x+1 & x-2 \\ -4 & 0 & 0 \\ 3 & -3 & 3 \end{vmatrix}\left(by\ R_3\to R_3-2R_1\right)=x\begin{vmatrix} 1 & 1 & 1 \\ -4 & 0 & 0 \\ 3 & -3 & 3 \end{vmatrix}+\begin{vmatrix} 0 & 1 & -2 \\ -4 & 0 & 0 \\ 3 & -3 & 3 \end{vmatrix}=xA+B$$

58. Line $L_1=\vec{r}=7\hat{i}+6\hat{j}+2\hat{k}+\lambda\left(-3\hat{i}+2\hat{j}+4\hat{k}\right)$ and line L_2 is

$L_2=\vec{r}=5\hat{i}+3\hat{j}+4\hat{k}+\mu\left(2\hat{i}+\hat{j}+3\hat{k}\right)$

As point C lies on L_1 and point D lies on L_2

$\Rightarrow \quad \overrightarrow{CD}=L_2-L_1=2\hat{i}+3\hat{j}-2\hat{k}+\lambda\left(-3\hat{i}+2\hat{j}+4\hat{k}\right)-\mu\left(2\hat{i}+\hat{j}+3\hat{k}\right)$ …(i)

Since it is parallel to $2\hat{i}-2\hat{j}-\hat{k}$

So, direction ratios will be proportional:

$\Rightarrow \quad \frac{2-3\lambda-2\mu}{2}=\frac{3+2\lambda-\mu}{-2}=\frac{-2+4\lambda-3\mu}{-1}$

On solving these equations, we get $\lambda=2, \mu=1$

Putting these value in (i),

$\Rightarrow \quad \overrightarrow{CD}=-6\hat{i}+6\hat{j}+3\hat{k} \quad \Rightarrow \quad \left|\overrightarrow{CD}\right|=9$

59. $\sin\cos^{-1}\left(\cos\left(\tan^{-1}x\right)\right)=p$

For $x\in R;\tan^{-1}x\in\left(\frac{-\pi}{2},\frac{\pi}{2}\right)$

$\cos\left(\tan^{-1}x\right)\in(0,1]$

$\cos^{-1}\cos\left(\tan^{-1}x\right)\in\left[0,\frac{\pi}{2}\right)$

$\sin\left[\cos^{-1}\cos\left(\tan^{-1}x\right)\right]\in[0,1)$

60.

$b_1+b_2=1\Rightarrow b_1(1+r)=1\Rightarrow b_1=\frac{1}{1+r}$

$\sum_{k=1}^{\infty}b_k=\frac{1}{(1+r)(1-r)}=\frac{1}{1-r^2}=2\Rightarrow r=\frac{-\sqrt{2}}{2}$

$b_1=\frac{1}{1+r}=\frac{1}{1-\frac{\sqrt{2}}{2}}=\left(2+\sqrt{2}\right)$

61. The hyperbola is

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$$\left(\frac{x-\sqrt{2}}{4}\right)^2-\frac{(y+\sqrt{2})^2}{2}=1$$

$a=2, b=\sqrt{2}, e=\sqrt{\frac{3}{2}}$.

If D be the centre, then

$CD = ae = \sqrt{6}$ and $DA = a = 2$

$\therefore$ $AC=\sqrt{6}-2$ and $BC=\frac{b^2}{a}=1$

Now area $\Delta ABC=\frac{1}{2}(AC)(BC)=\sqrt{\frac{3}{2}}-1$

62. Direction ratio of line

Let P(4, 3, 8), Q(3, 5, 7)

Now, direction ratio of PQ = –1, 2, –1

$\therefore PQ \perp L_1 \quad \Rightarrow \quad -l+6-4=0 \quad \Rightarrow \quad l=2$

Now, point Q (3, 5, 7) satisfies the line L_1

$\Rightarrow \quad a=1 \ \& \ b=3$

Now, shortest distance between

$L_1=\frac{x-1}{2}=\frac{y-2}{3}=\frac{z-3}{4}$ and $L_2=\frac{x-2}{3}=\frac{y-4}{4}=\frac{z-5}{5}$

Shortest distance = $d=\frac{\left(-\hat{i}+2\hat{j}-\hat{k}\right)\left(\hat{i}+2\hat{j}+2\hat{k}\right)}{\sqrt{1+3+1}}=\frac{1}{\sqrt{6}}$

63. $$I=\int_{-2}^{2}\frac{\left(1+\sqrt{|-x|-(-x)}3^{-x}\right)+\left(\sqrt{|-x|-(-x)}\right)3^{-(-x)}}{3^{-x}+3^{-(-x)}}$$

$$\Rightarrow I=\int_{-2}^{2}\frac{\left(1+\sqrt{|x|+x}\right)3^{-x}+\left(\sqrt{|x|+x}\right)3^{x}}{3^{x}+3^{x}}$$

$$\Rightarrow 2I=\int_{-2}^{2}\frac{\left(1+\sqrt{|x|+x}+\sqrt{|x|-x}\right)\left(3^{x}+3^{-x}\right)}{3^{x}+3^{-x}}dx$$

$$\Rightarrow 2I=\int_{-2}^{2}\left(1+\sqrt{|x|+x}+\sqrt{|x|-x}\right)dx$$

$$\Rightarrow 2I=2\int_{0}^{2}\left(1+\sqrt{|x|+x}+\sqrt{|x|-x}\right)dx$$

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$$\Rightarrow 2I = 2\int_0^2 \left(1+\sqrt{2x}+\sqrt{0}\right)dx$$

$$\Rightarrow I = \int_0^2 \left(1+\sqrt{2x}\right)dx = \left[x+\frac{2\sqrt{2}}{3}x^{3/2}\right]_0^2$$

$$\Rightarrow I = \frac{8}{3}+2$$

64. $\sqrt{3}\left[1+\sqrt{25}+\sqrt{81}+\sqrt{169}.....\right] = 435\sqrt{3}$

$\sqrt{3}\left[1+5+9+13.....T_n\right] = 435\sqrt{3}$

$\Rightarrow \sqrt{3}\times\frac{n}{2}\left[2+(n-1)4\right] = 435\sqrt{3}$

$4n^2-2n-870=0 \Rightarrow n=15$

65. $\alpha = \dfrac{\int_0^1 (1-x^{50})^{101}dx}{\int_0^1 (1-x^{50})^{100}dx} = \dfrac{I_2}{I_1}$

$I_2 = \int_0^1 \left(1-x^{50}\right)\left(1-x^{50}\right)^{100}dx$

$I_2 = I_1 - \int_0^1 x\cdot x^{49}\left(1-x^{50}\right)^{100}dx$

$I_2 = I_1 - \left[\dfrac{-x(1-x^{50})^{101}}{5050}\right]_0^1 - \int_0^1 \dfrac{(1-x^{50})^{101}}{5050}dx$

$I_2 = I_1 - \dfrac{I_2}{5050}$

$\Rightarrow \quad \alpha = \dfrac{I_2}{I_1} = \dfrac{5050}{5051}$

66.Even G = original signal is green

E_1 = A receives the signal correct

E_2 = B receives the signal correct

E = Signal received by B is green

P = (Signal received by B is green)

$= P\left(GE_1E_2\right)+P\left(G\overline{E}_1\overline{E}_2\right)+P\left(\overline{G}E_1\overline{E}_2\right)+P\left(\overline{G}\overline{E}_1E_2\right)$

$P(E) = \dfrac{46}{5\times 16}$

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$$P(G/E)=\frac{40/5\times16}{46/5\times16}=\frac{20}{23}$$

67. Given area = 56 sq. units

$$\begin{vmatrix} k & -3k & 1 \\ 5 & k & 1 \\ -k & 2 & 1 \end{vmatrix} = \pm 56$$

Expanding, we get

$k(k-2)-5(-3k-2)-k(-3k-k) = \pm 56$

$\Rightarrow k^2-2k+15k+10+3k^2+k^2=\pm 56$

$\Rightarrow 5k^2+13k+10=\pm 56$

Taking the positive sign

$5k^2+13k-46=0$

$\Rightarrow (5k+23)(k-2)=0$

$\therefore$ $k=2$ is an integer

Taking the negative sign

$5k^2+13k+66=0$

Here, $D=(13^2-4\times5\times66)<0$

$\Rightarrow$ There is no solution in this case.

So, the vertices are $A(2,-6)$, $B(5,2)$ and $C(-2,2)$.

The equation of altitude from A is $x=2$ and the equation of altitude from C is $y-2=-\frac{3}{8}(x+2)$

$\because$ $3x+8y-10=0$

Solving the two we get the orthocentre as $\left(2,\frac{1}{2}\right)$.

68. For real roots we get $q^4-4p^2\geq 0, \Rightarrow q^2\geq 2p$

If P = 1 $\Rightarrow$ q = 2, 3, 4, 5

If P = 2 $\Rightarrow$ q = 2, 3, 4, 5

If P = 3 $\Rightarrow$ q = 3, 4, 5

If P = 4 $\Rightarrow$ q = 3, 4, 5

If P = 5 $\Rightarrow$ q = 4, 5

Total number of different possibilities = 16

69. $\vec{u}=3\hat{i}-\hat{j},\vec{v}=2\hat{i}+\hat{j}-\lambda\hat{k}$

$$\Rightarrow \frac{\vec{u}.\vec{v}}{|\vec{u}||\vec{v}|}=\cos\theta$$

$$\Rightarrow \frac{5}{\sqrt{10}\sqrt{5+\lambda^2}}=\frac{\sqrt{5}}{2\sqrt{7}}$$

$$\Rightarrow \lambda^2=9\Rightarrow\lambda=3\left(\because \lambda>0\right)$$

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$$\vec{v} = \vec{v}_1 + \vec{v}_2$$

$$\Rightarrow \left|\vec{v}\right|^2 = \vec{v}_1^2 + \vec{v}_2^2 + 2\vec{v}_1.\vec{v}_2$$

$$\Rightarrow 14 = v_1^2 + v_2^2 + 0 \qquad \left(\because \vec{v}_1 \perp \vec{v}_2\right)$$

$$\Rightarrow \left|\vec{v}_1^2\right| + \left|\vec{v}_1^2\right| = 14$$

70. For continuity

$$k = f(0) = \lim_{x\to 0} f(x) = \lim_{x\to 0} \frac{e^{\tan x} - e^x}{\tan x - x} + \lim_{x\to 0} \frac{\ln(\sec x + \tan x) - x}{\left(\frac{\tan x - x}{x^3}\right)\cdot x^3}$$

$$= \lim_{x\to 0} \frac{e^x\left(e^{\tan x - x} - 1\right)}{\tan x - x} + 3\lim_{x\to 0} \frac{\ln(\sec x + \tan x) - x}{x^3}$$

$$= 1 + 3\lim_{x\to 0} \frac{\sec x - 1}{3x^2} = 1 + \frac{1}{2} = \frac{3}{2}$$

71. A function $f : A \to B$ such that $f(0) \le f(1) \le f(2)$

In the following cases

$$case\ 1 : f(0) < f(1) < f(2) \to {}^8C_3$$

$$case\ 2 : f(0) = f(1) = f(3) \to {}^8C_1$$

$$case\ 3 : f(0) = f(1) < f(2) \to {}^8C_2$$

$$Case\ 4 : f(0) \angle f(1) = f(2) \to {}^8C_2$$

$$\therefore {}^8C_3 + {}^8C_2 + {}^{-}8C_2 + {}^{-8}C_1$$

$$= {}^9C_3 + C_{C_2} = {}^{10}C_3$$

72. $A = \{6, 11, 16, 21, 26, 31, 36, 41, 46, 51, 56, 61, 66, 71, 76, 81, 86, 91, ...\}$

$B = \{16, 23, 30, 37, 44, 51, 58, 65, 72, 79, 86, 93, 100, ...\}$

$A \cap B = \{16, 51, 86, ...\}$

For set $'A' \Rightarrow T_{2024} = 6 + (2024 - 1)(5) = 10121$

For set $'B' \Rightarrow T_{2024} = 16 + (2024 - 1)(7) = 14177$

So, for $(A \cap B) \Rightarrow T_n = 16 + (n-1)(35) \le 10121$

$$(n-1) \le \frac{10121 - 16}{35} = 288.71$$

$$n \le 289.71 \Rightarrow n = 289$$

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$\therefore n(A \cup B) = n(A) + n(B) - n(A \cap B)$

$= 2024 + 2024 - 289 = 3759$

73. The coefficient of x^6 in the given expression = coefficient of x^6 in

$(1+{}^6C_1x^6)(1+{}^5C_1x^5)(1+{}^4C_1x^4)(1+{}^3C_1x^3+{}^3C_2x^6)(1+{}^2C_1x^2+{}^2C_2x^4)(1+x)$

= coefficient of x^6 in $(1+6x^6+5x^5+4x^4)(1+2x^2+3x^3+x^4+6x^5+3x^6)(1+x)$

= coefficient of x^6 in $(11x^5+17x^6)(1+x)$

74. $f''(x) = \lambda(x-1)$

$f(x) = \frac{\lambda x^2}{2} - \lambda x + C \Rightarrow f'(-1) = 0 \Rightarrow c = \frac{-3\lambda}{2}$

$f(x) = \frac{\lambda x^3}{6} - \frac{\lambda x^2}{2} - \frac{3\lambda}{2}x + d$

$f(1) = -6 \Rightarrow -11\lambda + 6d 0 - 36$ …(i)

$f(-1) = 10 \Rightarrow 5\lambda + 6d - 60$ …(ii)

From (i) & (ii) λ=6 & d = 5

$f(x) = x^3 - 3x^2 - 9x + 5$

Which has minimum at x = 3

75. Let $y = f(f(f(x))) + (f(x))^2$

$\frac{dy}{dx} = f'(f(f(x))) \cdot f'(f(x)) \cdot f'(x) + 2f(x)f'(x) = 33$

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